Reg. No									

St. Joseph's College of Arts & Science (Autonomous)
St. Joseph's College Road, Cuddalore – 607001
B.Sc Degree Examination - APRIL 2021

CS306 – FUNDAMENTALS OF ALGORITHMS

Time : 3 hrs **Max Marks :75**

SECTION – A (5X5=25)

Answer any **FIVE** Questions

1. Explain the binary search algorithm with example.
2. Explain the maximum minimum algorithm with example.
3. What is Multi Stage Graph? Explain.
4. Briefly explain the Depth First Search and traversal technique.
5. Describe the general method of Graph colorings.
6. What is 0/1 knapsack problem? Explain.
7. Discuss about the Non-deterministic algorithm.
8. Briefly discuss Divide and conquer.

SECTION – B (5X10=50)

Answer any **FIVE** Questions

9. Explain with example about the complexity analysis.
10. Explain the Merge Sort method with example.

11. Discuss about the Travelling Salesman problem in Dynamic Programming.

12. Describe the Breadth First search and traversal technique with exapmple.

13. State and explain backtracking algorithm, with an Example.

14. Explain the general method of Greedy algorithm.

15. Explain the algorithm with example to find the shortest path.

16. Explain the classes Np-Hard and Np-Complete.

****